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Questions on Data Types, Variables and Arrays

Integer and Floating Data Types

1. What is the range of short data type in Java?

a) -128 to 127

b) -32768 to 32767

Answer : b) -32768 to 32767

c) -2147483648 to 2147483647

d) None of the mentioned

2. What is the range of byte data type in Java?

a) -128 to 127

Answer : a) -128 to 127

b)

c) -32768 to 32767

d) -2147483648 to 2147483647

e) None of the mentioned

3. Which of the following are legal lines of Java code?

1. int w = (int)888.8;

2. byte x = (byte)100L;

3. long y = (byte)100;

4. byte z = (byte)100L;

a) 1 and 2

b) 2 and 3

c) 3 and 4

d) All statements are correct

Answer : d) All statements are correct

4. An expression involving byte, int, and literal numbers is promoted to which of these?

a) Int

Answer : a) int

b) long

c) byte

d) float

5. Which of these literals can be contained in float data type variable?

a) -1.7e+308

b) -3.4e+038

Answer : b) -3.4e+038

c) +1.7e+308

d) -3.4e+050

6. Which data type value is returned by all transcendental math functions?

a) int

b) Float

Answer : b) double

- c) Double
- d) long

7. What will be the output of the following Java code?

```
class average {  
    public static void main(String args[])  
    {  
        double num[] = {5.5, 10.1, 11, 12.8, 56.9, 2.5};  
        double result;  
        result = 0;  
        for (int i = 0; i < 6; ++i)  
            result = result + num[i];  
        System.out.print(result/6);  
    }  
}
```

- a) 16.34
- b) 16.566666644
- c) 16.46666666666667
- d) 16.46666666666666

Answer : c) 16.46666666666667

8. What will be the output of the following Java statement?

```
class output {  
    public static void main(String args[])  
    {  
        double a, b, c;  
        a = 3.0/0;  
        b = 0/4.0;  
        c = 0/0.0;  
        System.out.println(a);  
        System.out.println(b);  
        System.out.println(c);  
    }  
}
```

- a) Infinity
- b) 0.0
- c) NaN

d) all of the mentioned

Answer : d) all of the mentioned

9. What will be the output of the following Java code?

```
class increment {  
    public static void main(String args[])  
    {  
        int g = 3;  
        System.out.print(++g * 8);  
    }  
}
```

- a) 25
- b) 24

- c) 32
- d) 33

Answer : c) 32

10. What will be the output of the following Java code?

```
class area {  
    public static void main(String args[])  
    {  
        double r, pi, a;  
        r = 9.8;  
        pi = 3.14;  
        a = pi * r * r;  
        System.out.println(a);  
    }  
}
```

- a) 301.5656
- b) 301
- c) 301.56
- d) 301.56560000

Answer : a) 301.5656

Character and Boolean Data Types

11. What is the numerical range of a char data type in Java?

- a) -128 to 127
- b) 0 to 256
- c) 0 to 32767
- d) 0 to 65535

Answer : d) 0 to 65535

12. Which of these coding types is used for data type characters in Java?

- a) ASCII
- b) ISO-LATIN-1
- c) UNICODE
- d) None of the mentioned

Answer : c) UNICODE

13. Which of these values can a boolean variable contain?

- a) True & False
- b) 0 & 1
- c) Any integer value
- d) true

Answer : a) True & False

14. Which of these occupy first 0 to 127 in Unicode character set used for characters in Java?

- a) ASCII
- b) ISO-LATIN-1
- c) None of the mentioned

Answer : a) ASCII

d) ASCII and ISO-LATIN1

15. Which one is a valid declaration of a boolean?

- a) boolean b1 = 1;
- b) boolean b2 = 'false';
- c) boolean b3 = false;
- d) boolean b4 = 'true'

Answer : c) boolean b3 = false;

16. What will be the output of the following Java program?

```
class array_output {  
    public static void main(String args[])  
    {  
        char array_variable [] = new char[10];  
        for (int i = 0; i < 10; ++i)  
        { array_variable[i] = 'i';  
          System.out.print(array_variable[i] + "" );  
          i++;  
        }  
    }  
}
```

- a) i i i i i
- b) 0 1 2 3 4
- c) i j k l m
- d) None of the mentioned

Answer : a) i i i i i

17. What will be the output of the following Java program?

```
class mainclass {  
    public static void main(String args[])  
    {  
        char a = 'A';  
        a++;  
        System.out.print((int)a);  
    }  
}
```

- a) 66
- b) 67
- c) 65
- d) 64

Answer : a) 66

18. What will be the output of the following Java program?

```
class mainclass {  
    public static void main(String args[])  
    {  
        boolean var1 = true;  
        boolean var2 = false;  
        if (var1)  
            System.out.println(var1);  
        else
```

```

        System.out.println(var2);
    }
}
a) 0
b) 1

```

c) True Answer : c) true
d) false

19. What will be the output of the following Java code?

```

class booloperators {
    public static void main(String args[])
    {
        boolean var1 = true;
        boolean var2 = false;
        System.out.println((var1 & var2));
    }
}
a) 0
b) 1

```

c) true
d) false Answer : d) false

20. What will be the output of the following Java code?

```

class asciicodes {
    public static void main(String args[])
    {
        char var1 = 'A';
        char var2 = 'a';
        System.out.println((int)var1 + " " + (int)var2);
    }
}

```

a) 162
b) 65 97 Answer : b) 65 97
c) 67 95
d) 66 98

21. What is the order of variables in Enum?

a) Ascending order
b) Descending order
c) Random order
d) Depends on the order() method Answer : d) Depends on the order() method

22. Can we create an instance of Enum outside of Enum itself?

a) True
b) False Answer : b) False

23. What will be the output of the following Java code?

```
enum Season
{
    WINTER, SPRING, SUMMER, FALL
};
System.out.println(Season.WINTER.ordinal());
```

a) 0

Answer : a) 0

b) 1

c) 2

d) 3

24. If we try to add Enum constants to a TreeSet, what sorting order will it use?

a) Sorted in the order of declaration of Enums

b) Sorted in alphabetical order of Enums
order of Enums

Answer : b) Sorted in alphabetical

c) Sorted based on order() method

d) Sorted in descending order of names of Enums

25. What will be the output of the following Java code snippet?

```
class A
{
}
enum Enums extends A
{
    ABC, BCD, CDE, DEF;
}
```

a) Runtime Error

b) Compilation Error

Answer : b) Compilation Error

c) It runs successfully

d) EnumNotDefined Exception

26. What will be the output of the following Java code snippet?

```
enum Levels
{
    private TOP,

    public MEDIUM,

    protected BOTTOM;
}
```

a) Runtime Error

b) EnumNotDefined Exception

c) It runs successfully

d) Compilation Error

Answer : d) Compilation Error